FINAL-YEAR COMPUTER ENGINEERING · SIIT, THAMMASAT UNIVERSITY

Singha Junchan

สิงหา จุลจันทร์ / Pathum Thani, Thailand

singhaju.github.io
linkedin.com/in/singha-junchan
github.com/singhaju

I build access control for cloud and AI systems.

Open to: security internships from January 2027, in Thailand or abroad · full-time roles from May 2027

COMPETITIVE PROGRAMMING
#1 in Thailand
TOI18 national champion, 2022, after 1,000+ hours of training

CLIENT PROJECT
TU Alert
built for **Thammasat University**; **I lead the team**; security built in; launching soon

RESEARCH
9 research projects
2 accepted (one **Best Paper**), 3 submitted, 4 in preparation; first author on 8

INDUSTRY
2 internships
Allianz Ayudhya (security governance and risk, with the CISO) and **Amity Solutions** (LLM chatbots)

LEADERSHIP AND TEACHING
11 student events
led SIIT First Meet 2024 (about 500 people) and organised 10 more; founded **two coaching platforms**

CERTIFICATION
ISC2 Certified
Certified in Cybersecurity (CC), 2025; CompTIA Security+ planned Dec 2026

01 Experience

knowledge | tech used

Team Lead and Developer, TU Alert

Launching soon

Thammasat University (client project) 2026

- Campus incident reporting **inside LINE**, built for Thammasat University (launching soon). **I lead the team** and write code.
- **LINE Official Account, group bot and two LIFF apps**: students report with photo and map pin, track and chat; staff accept, transfer and resolve from a phone or the portal. **Security built in**: roles, second-step sign-in, audit log, PDPA erasure.

Secure web development · Role-based access control · Data retention and erasure
TypeScript · Next.js · Hono · Bun · Prisma · MariaDB

Information Security Intern

Allianz Ayudhya Jun to Jul 2026

- Learned **security governance and risk** from the CISO, and helped design how the team's **third-party risk** work uses AI.
- Built a **questionnaire agent** (evidence, a reason and a **confidence score** per answer; **gap and fit-gap analysis**) and a **guidance agent** that checks inputs, suggests next steps and flags risks. Tested on real, anonymized cases.

Information security governance · Risk management · Third-party risk management

Founder, Curriculum Author and Lead Instructor

TOI Tutor and ABIG Center (online) 2022 to present

- **Founded two olympiad coaching platforms** (2022); coached hundreds of students.
- Wrote **COM300** (C++ and algorithms, **128 recorded lessons**) and **COM100** (maths and Python).

Algorithms and data structures · Competitive programming · Curriculum design
C++ · Python

R&D Intern, AI research team

Amity Solutions Jul to Aug 2023

- Read **LLM research papers**, applied their ideas and **tested them end to end**.
- Built **RAG chatbots**: search the knowledge base, give the best documents to the LLM, return the answer.

Large language models · Retrieval-augmented generation (RAG) · Applying research papers

02 Research

9 research projects: 2 accepted, 3 submitted, 4 in preparation. First author on 8.

FEDAC

First author of 4 · Under review · IEEE TDSC

Authorization-as-Result: decide once, then a short-lived token per service. **Risk-scored sign-in** up to a zero-knowledge proof, many policy types compiled into one template, blockchain-anchored policies.

- **Risk-adaptive sign-in:** the more sensitive the service, the stronger the proof, in four tiers from single sign-on to a **zero-knowledge proof**; a **risk score** from device, location and behaviour must stay under a threshold.

RESULT 1.4 ms median authorization latency on AWS EKS, about 7,100 requests per second (paper under review)

Zero trust · Zero-knowledge proofs · Service mesh · Go · Rust · Groth16 zk-SNARK · Envoy + Istio · IPFS · AWS EKS

U-HAP

Second author of 4 · presented the paper · Presented · IEEE JCSSE 2026 · Best Paper Award

Authorization as a compiler problem: RBAC, ABAC, ACL and deny rules in **one graph**, compiled into indexes; every decision is deterministic (deny wins).

- **One model for every rule type:** RBAC, ABAC, ACL, role hierarchy and **deny rules** share one meaning, without rewriting policies into a standard.

RESULT ~2.7× faster policy updates reach a first decision vs OPA at 2,000 policies, with identical decisions

Access control (RBAC, ABAC, ACL) · Kubernetes admission · Policy compilation · Python · Kubernetes webhook · OPA (baseline)

ZKAudit

Accepted · KST 2027

First author. **Prove you may audit without revealing your credential**, bound to the exact log state you check; sharded proof checks and Merkle-chained logs.

Zero-knowledge proofs · Cloud audit logs · Merkle commitments · Rust · Groth16 zk-SNARKs · AWS EC2

Post-quantum authorization across Kubernetes clusters

Submitted

First author. Each cluster **verifies that a decision used every allow and deny rule**, under **post-quantum** signatures (ML-DSA, ML-KEM); the heavy check is paid once per session.

Post-quantum cryptography · Multi-cluster Kubernetes · Authorization · Go · ML-DSA · ML-KEM · Hyperledger Fabric · AWS EKS

Also submitted: Self-healing fog computing for industrial IoT | **Also in preparation:** MoCA: AI containment · ZK-ALBA · Threshold-gated encrypted SIEM · Fresh answers from elastic multi-cloud services

03 Education

B.Eng. in Computer Engineering

Aug 2023 to May 2027 (expected)

Minor in Cloud Computing and Cybersecurity. Sirindhorn International Institute of Technology (SIIT), Thammasat University.

GPA 3.67 / 4.00, first-class honours expected. Full scholarship: 4-Year Continuing Academic Outstanding Student Program.

04 Awards

2022 **National Champion: 1st place in Thailand and Gold Medal** Thailand Olympiad in Informatics (TOI18)

2026 **Best Paper Award** IEEE JCSSE 2026, Bangkok, for U-HAP (second author of 4)

2025 **1st place overall** Image Recognition Programming Challenge, Osaka Institute of Technology and SIIT

Skills **Main: Python · C++** | **Also:** Go · Rust · TypeScript · C · Java · Bash · SQL | **Cloud and security:** Kubernetes · AWS · Docker · Envoy and Istio
Projects TU Alert · Novel Nest · Arduino retro console · Automated Cat Feeder · Velour Mall

REFERENCES

Apichai Wongchindakhun Chief Information Security Officer (CISO), Allianz Ayudhya LinkedIn
Prof. Somchart Fugkeaw Professor, School of ICT, SIIT, Thammasat University somchart@siit.tu.ac.th
Pachara Co-developer, TU Alert, SIIT, Thammasat University Contact through me